

```
Microsoft Windows [Version 10.0.26200.8655]
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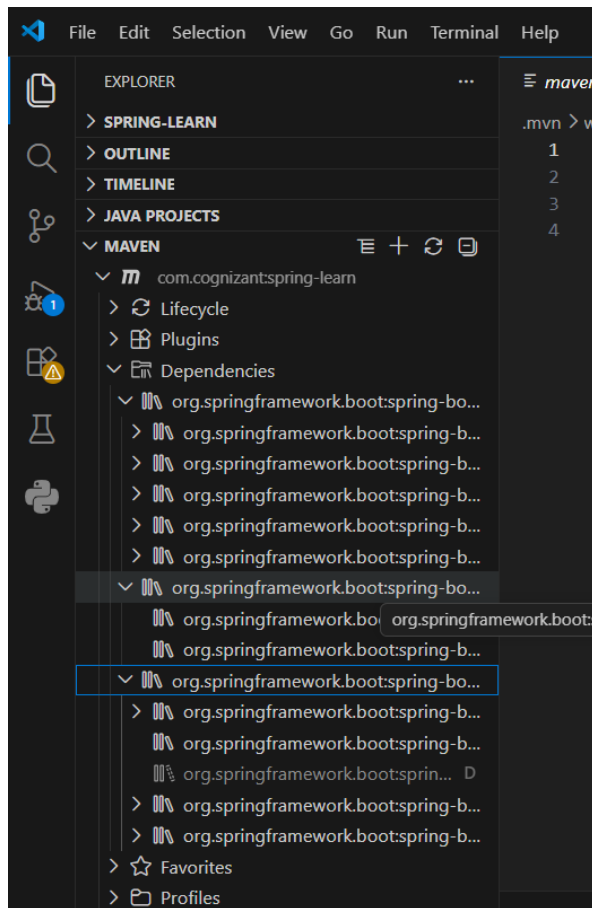
C:\Users\Khushi_Ahi\OneDrive\Desktop\SpringBoot\spring-learn> cmd /c ""C:\Users\Khushi_Ahi\.vscode\extensions\redhat.j
ava-1.55.0-win32-x64\jre17.0.11-win32-x86_64\bin\java.exe"" @C:\Users\Khushi_Ahi\AppData\Local\Temp\cp_5pojof964mfilwso
wiki2izf.argfile com.cognizant.spring_learn.SpringLearnApplication "
```

The image shows the Spring Boot logo, which consists of a stylized green 'S' and 'B' inside a circle, followed by the word 'Spring' in a green serif font and 'Boot' in a green sans-serif font, all enclosed within a green rectangular border.

```

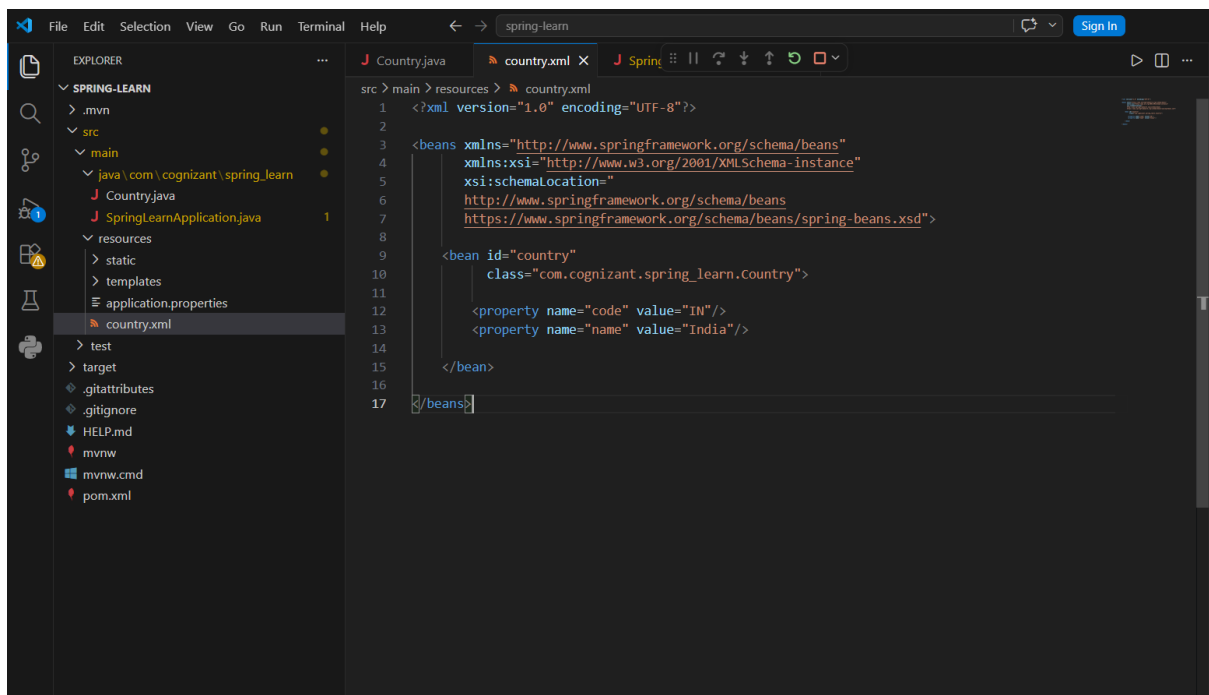
:: Spring Boot ::                                (v4.1.0)

2026-07-04T18:19:20.851495:30 INFO 13992 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication
: Starting SpringLearnApplication using Java 21.0.11 with PID 13992 (C:\Users\Khushi_Ahi\OneDrive\Desktop\SpringBoot
\spring-learn\target\classes started by Khushi_Ahi in C:\Users\Khushi_Ahi\OneDrive\Desktop\SpringBoot\spring-learn)
2026-07-04T18:19:20.856495:30 INFO 13992 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication
: No active profile set, falling back to 1 default profile: "default"
2026-07-04T18:19:20.936495:30 INFO 13992 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcesso
r: DevTools property defaults active Set [spring.devtools.add-properties-to: 'false' to disable
2026-07-04T18:19:20.937495:30 INFO 13992 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcesso
r: For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-04T18:19:22.112495:30 INFO 13992 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer
: Tomcat initialized with port 8080 (http)
2026-07-04T18:19:22.151495:30 INFO 13992 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService
: Starting service [Tomcat]
2026-07-04T18:19:22.151495:30 INFO 13992 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine
: Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-04T18:19:22.211495:30 INFO 13992 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContext: Initializ
e
r: Root WebApplicationContext: Initialization completed in 1272 ms
2026-07-04T18:19:22.817495:30 INFO 13992 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer
: Tomcat started on port 8080 (http) with context path '/'
2026-07-04T18:19:22.830495:30 INFO 13992 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication
: Started SpringLearnApplication in 2.61 seconds (process running for 3.222)
```



Exercise 4: Spring Core – Load Country from Spring Configuration XML

CODE

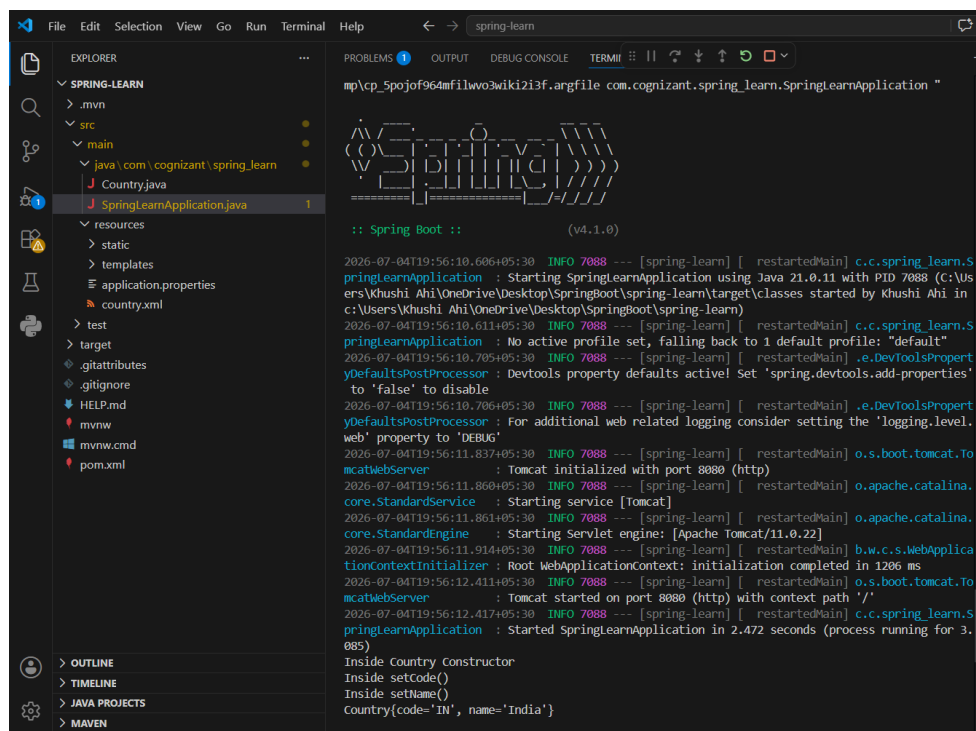


```
src > main > java > com > cognizant > spring_learn > J Country.java > Country
1 package com.cognizant.spring_learn;
2
3 public class Country {
4
5     private String code;
6     private String name;
7
8     public Country() {
9         System.out.println("Inside Country Constructor");
10    }
11
12    public String getCode() {
13        System.out.println("Inside getCode()");
14        return code;
15    }
16
17    public void setCode(String code) {
18        System.out.println("Inside setCode()");
19        this.code = code;
20    }
21
22    public String getName() {
23        System.out.println("Inside getName()");
24        return name;
25    }
26
27    public void setName(String name) {
28        System.out.println("Inside setName()");
29        this.name = name;
30    }
31
32    @Override
33    public String toString() {
34        return "Country{" +
35            "code=" + code + '\'' +
36            ", name=" + name + '\'' +
37            '}';
38    }
39 }
```

```
37
38 }
39 }
```

```
src > main > java > com > cognizant > spring_learn > J SpringLearnApplication.java > SpringLearnApplication
1 package com.cognizant.spring_learn;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 import org.springframework.context.ApplicationContext;
7 import org.springframework.context.support.ClassPathXmlApplicationContext;
8
9 @SpringBootApplication
10 public class SpringLearnApplication {
11
12     public static void displayCountry() {
13
14         ApplicationContext context =
15             new ClassPathXmlApplicationContext("country.xml");
16
17         Country country =
18             context.getBean("country", Country.class);
19
20         System.out.println(country.toString());
21     }
22
23     Run | Debug
24     public static void main(String[] args) {
25
26         SpringApplication.run(SpringLearnApplication.class, args);
27
28         displayCountry();
29     }
30 }
```

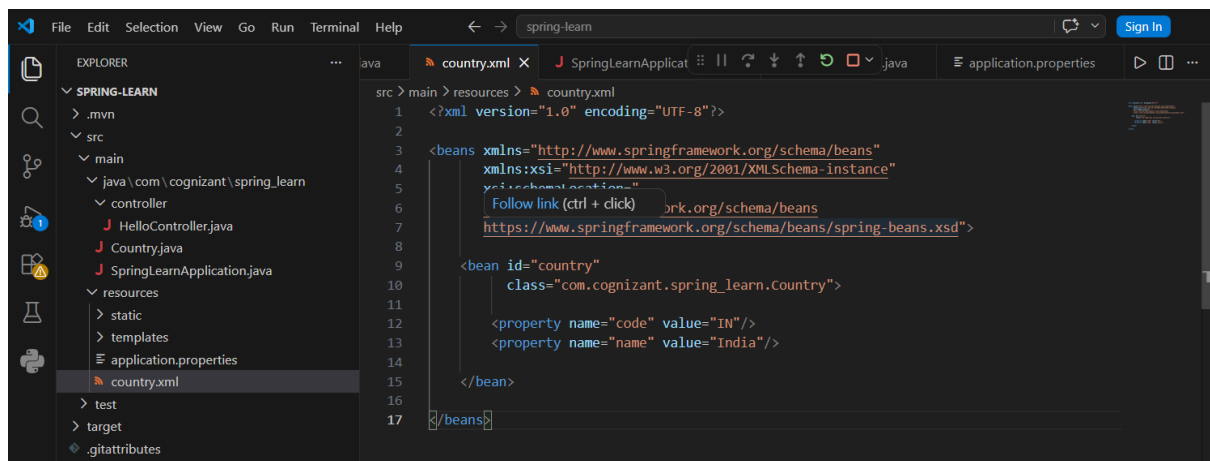
OUTPUT

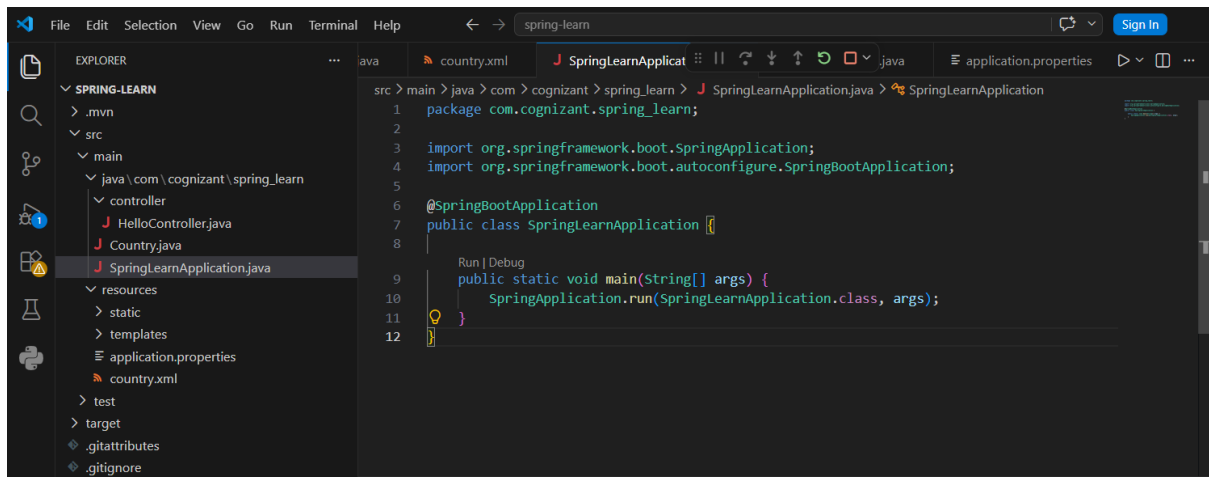


File- 2.spring-rest-handson

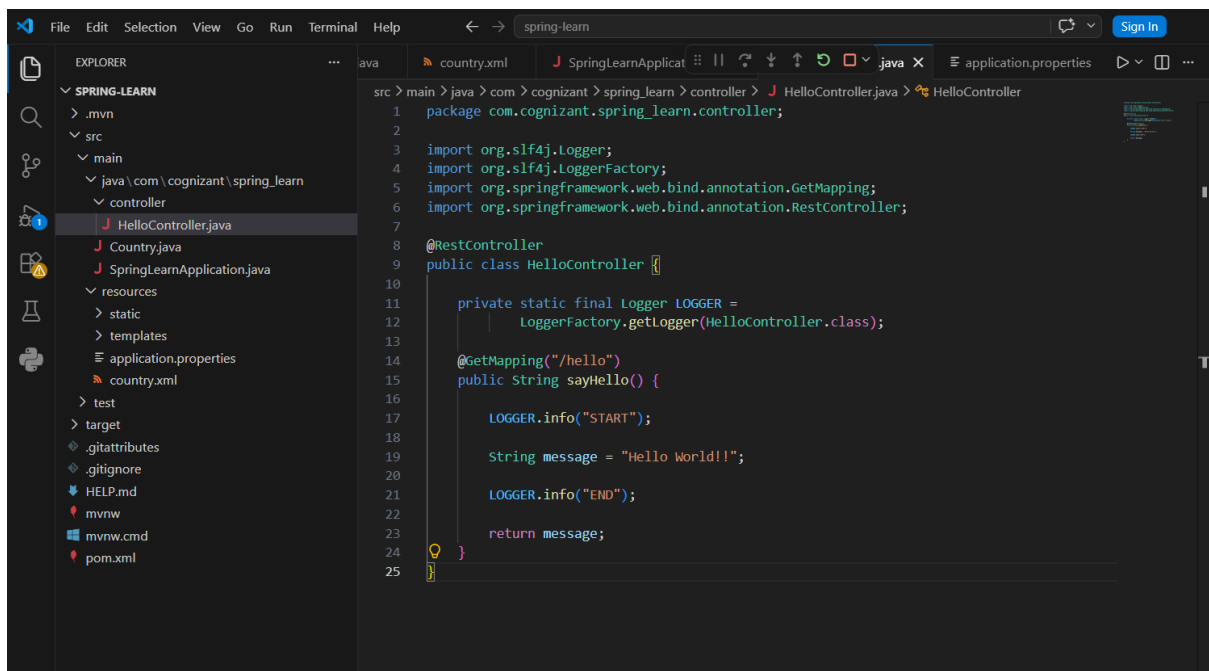
Exercise 1: Hello World RESTful Web Service

CODE

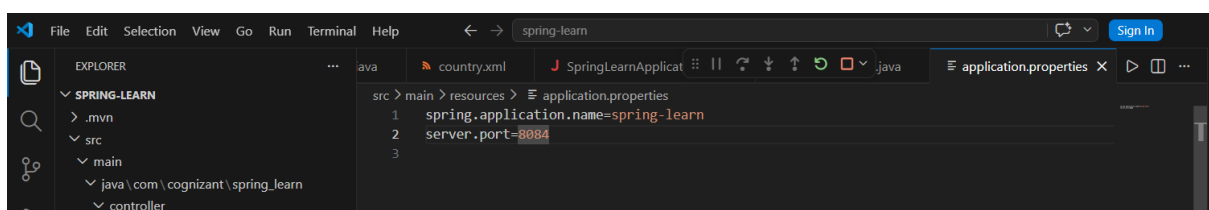




```
src > main > java > com > cognizant > spring_learn > SpringLearnApplication.java > SpringLearnApplication
1 package com.cognizant.spring_learn;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class SpringLearnApplication {
8
9     Run | Debug
10    public static void main(String[] args) {
11        SpringApplication.run(SpringLearnApplication.class, args);
12    }
13 }
```



```
src > main > java > com > cognizant > spring_learn > controller > HelloController.java > HelloController
1 package com.cognizant.spring_learn.controller;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.web.bind.annotation.GetMapping;
6 import org.springframework.web.bind.annotation.RestController;
7
8 @RestController
9 public class HelloController {
10
11    private static final Logger LOGGER =
12        LoggerFactory.getLogger(HelloController.class);
13
14    @GetMapping("/hello")
15    public String sayHello() {
16
17        LOGGER.info("START");
18
19        String message = "Hello World!!";
20
21        LOGGER.info("END");
22
23        return message;
24    }
25 }
```



```
src > main > resources > application.properties
1 spring.application.name=spring-learn
2 server.port=8084
3
```

OUTPUT

```
File Edit Selection View Go Run Terminal Help
spring-learn

EXPLORER
  SPRING-LEARN
    .mvn
    src
      main
        java\com\cognizant\spring_learn
          controller
            HelloController.java
            Country.java
            SpringLearnApplication.java
        resources
          static
          templates
        application.properties
        country.xml
    test
    target
    .gitattributes
    .gitignore
    HELP.md
    mvnw
    mvnw.cmd
    pom.xml

OUTLINE
TIMELINE
JAVA PROJECTS
MAVEN

Microsoft windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

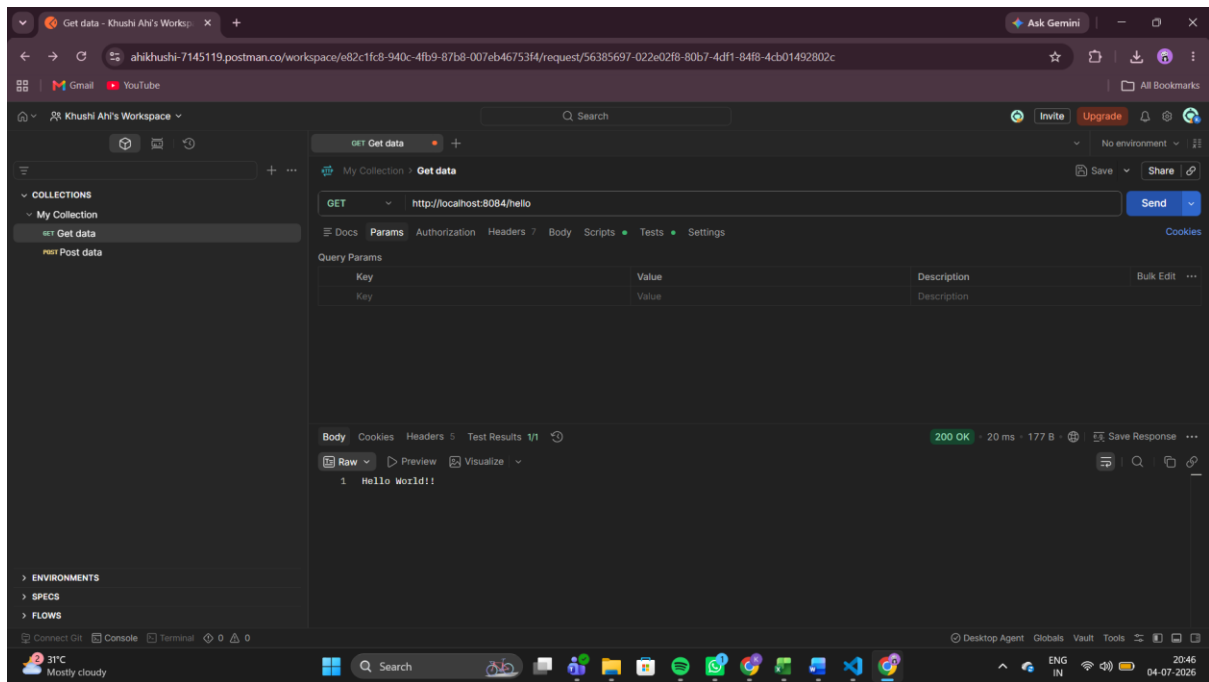
C:\Users\Khushi Ahi\OneDrive\Desktop\SpringBoot\spring-learn> cmd /c ""C:\Users\Khushi Ahi\.vscode\extensions\redhat.java-1.55.0-win32-x64\jre\21.0.11-win32-x86_64\bin\java.exe" @C:\Users\KHUSHI~1\AppData\Local\Temp\cp_5pojof964mfilmwo3wiki2i3f.argfile com.cognizant.spring_learn.SpringLearnApplication ""

:: Spring Boot :: (v4.1.0)

2026-07-04T20:27:32.249+05:30 INFO 24968 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 21.0.11 with PID 24968 (C:\Users\Khushi Ahi\OneDrive\Desktop\SpringBoot\spring-learn\target\classes started by Khushi Ahi i n C:\Users\Khushi Ahi\OneDrive\Desktop\SpringBoot\spring-learn)
2026-07-04T20:27:32.254+05:30 INFO 24968 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-04T20:27:32.338+05:30 INFO 24968 --- [spring-learn] [ restartedMain] .e.DevToolsProper tyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties ' to 'false' to disable
2026-07-04T20:27:32.338+05:30 INFO 24968 --- [spring-learn] [ restartedMain] .e.DevToolsProper tyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level .web' property to 'DEBUG'
2026-07-04T20:27:33.531+05:30 INFO 24968 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.T omcatWebServer : Tomcat initialized with port 8084 (http)
2026-07-04T20:27:33.559+05:30 INFO 24968 --- [spring-learn] [ restartedMain] o.apache.catalina .core.StandardService : Starting service [Tomcat]
2026-07-04T20:27:33.560+05:30 INFO 24968 --- [spring-learn] [ restartedMain] o.apache.catalina .core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-04T20:27:33.613+05:30 INFO 24968 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplic ationContextInitializer : Root WebApplicationContext: initialization completed in 1273 ms
2026-07-04T20:27:34.118+05:30 INFO 24968 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.T omcatWebServer : Tomcat started on port 8084 (http) with context path '/'
2026-07-04T20:27:34.127+05:30 INFO 24968 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 2.529 seconds (process running for 3
```

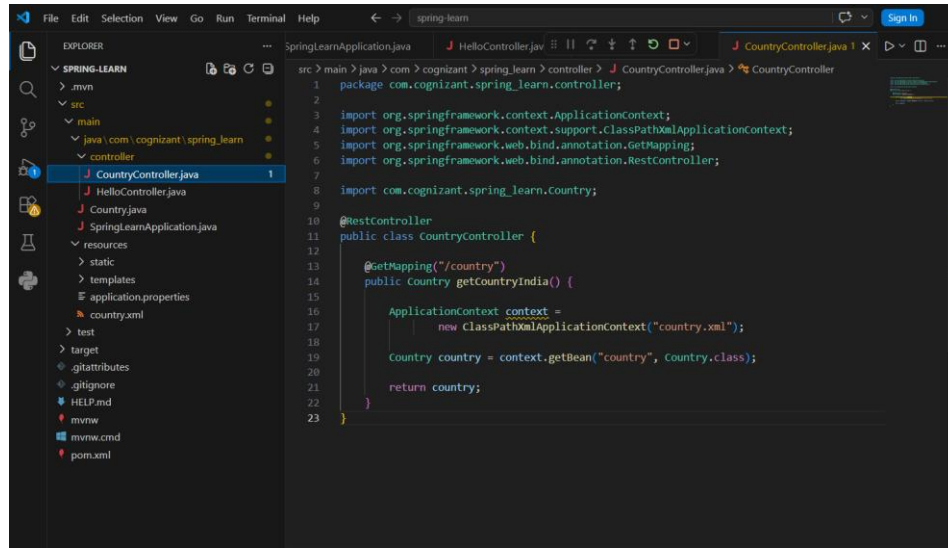
```
SpringLearnApplication : Started SpringLearnApplication in 2.529 seconds (process running for 3
.159)
2026-07-04T20:29:24.004+05:30 INFO 24968 --- [spring-learn] [nio-8084-exec-1] o.a.c.c.C.[Tomcat
].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-04T20:29:24.005+05:30 INFO 24968 --- [spring-learn] [nio-8084-exec-1] o.s.web.servlet.D
ispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-04T20:29:24.009+05:30 INFO 24968 --- [spring-learn] [nio-8084-exec-1] o.s.web.servlet.D
ispatcherServlet : Completed initialization in 2 ms
2026-07-04T20:29:24.076+05:30 INFO 24968 --- [spring-learn] [nio-8084-exec-1] c.c.s.controller.
HelloController : START
2026-07-04T20:29:24.076+05:30 INFO 24968 --- [spring-learn] [nio-8084-exec-1] c.c.s.controller.
HelloController : END
```

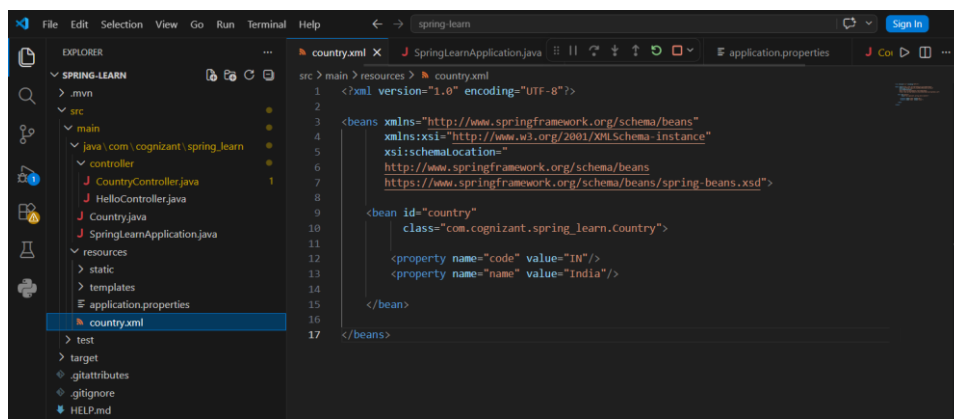
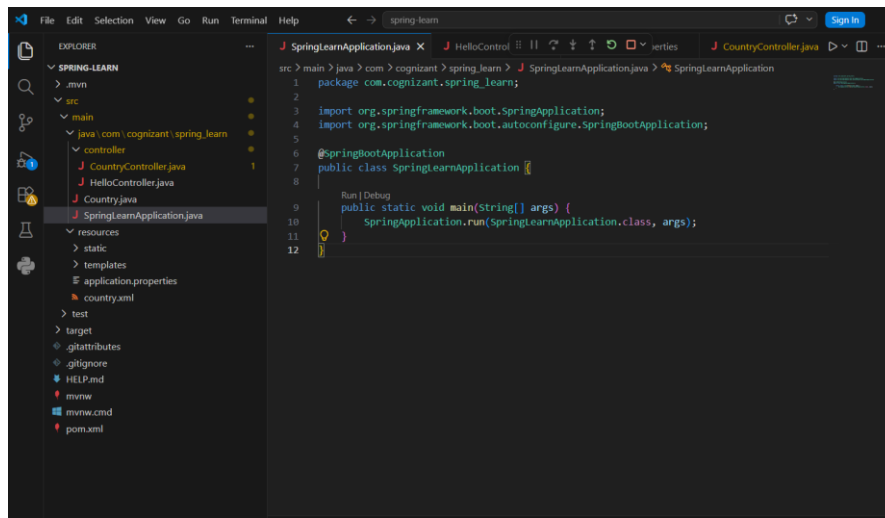




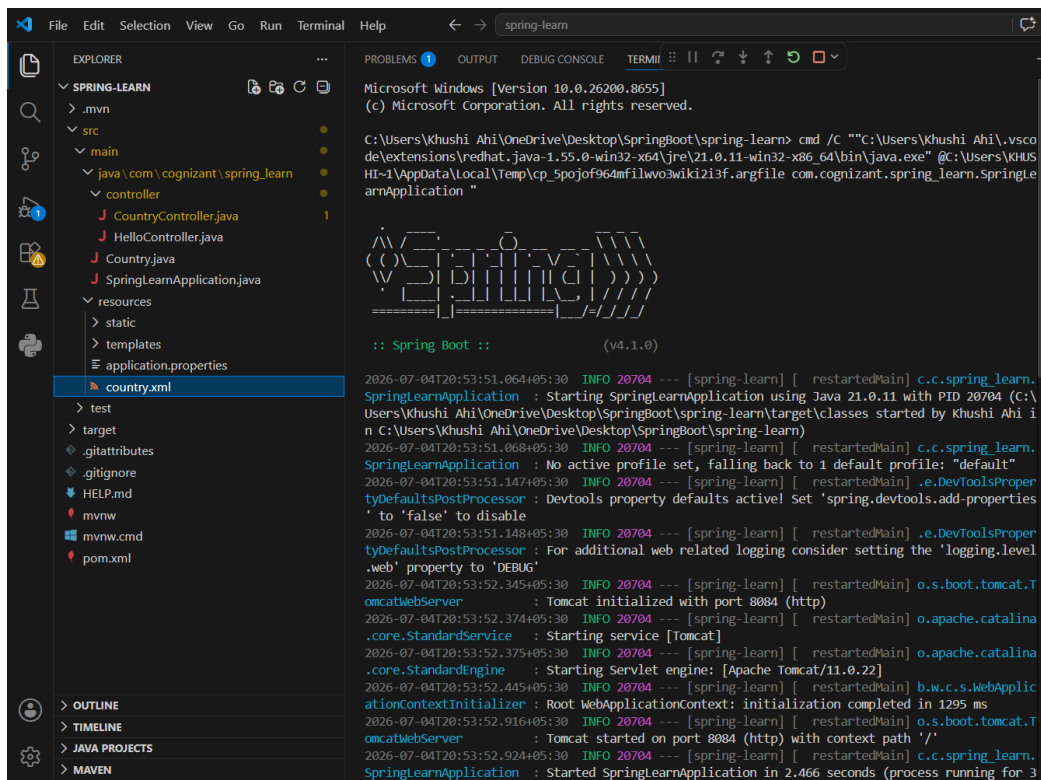
Exercise 2: REST - Country Web Service

CODE





OUTPUT




```

SpringLearnApplication : Started SpringLearnApplication in 2.466 seconds (process running for 3
.1)
2026-07-04T20:54:05.599+05:30 INFO 20704 --- [spring-learn] [nio-8084-exec-1] o.a.c.c.C.[Tomcat
].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-04T20:54:05.600+05:30 INFO 20704 --- [spring-learn] [nio-8084-exec-1] o.s.web.servlet.D
ispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-04T20:54:05.604+05:30 INFO 20704 --- [spring-learn] [nio-8084-exec-1] o.s.web.servlet.D
ispatcherServlet : Completed initialization in 3 ms
Inside Country Constructor
Inside setCode()
Inside setName()
Inside getCode()
Inside getName()
Inside Country Constructor
Inside setCode()
Inside setName()
Inside getCode()
Inside getName()

```

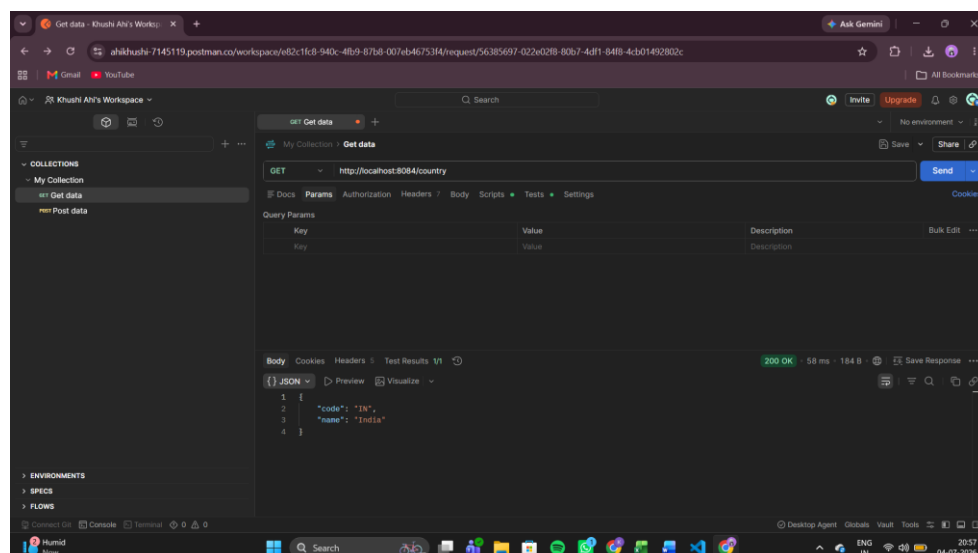


localhost:8084/country

localhost:8084/country

Pretty print

```
{"code":"IN","name":"India"}
```



Get data - Khushi Ah's Worksp...

ahikhushi-7145119 postman.co/workspace/e82c1c9-940c-48b9-87b8-007eb4675314/request/56385697-022e028-80b7-4d81-848b-4c1d1492802c

My Collection > Get data

GET http://localhost:8084/country

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body

JSON

```

1 {
2   "code": "IN",
3   "name": "India"
4 }

```

200 OK 58 ms 184 B

ENVIRONMENTS

SPECS

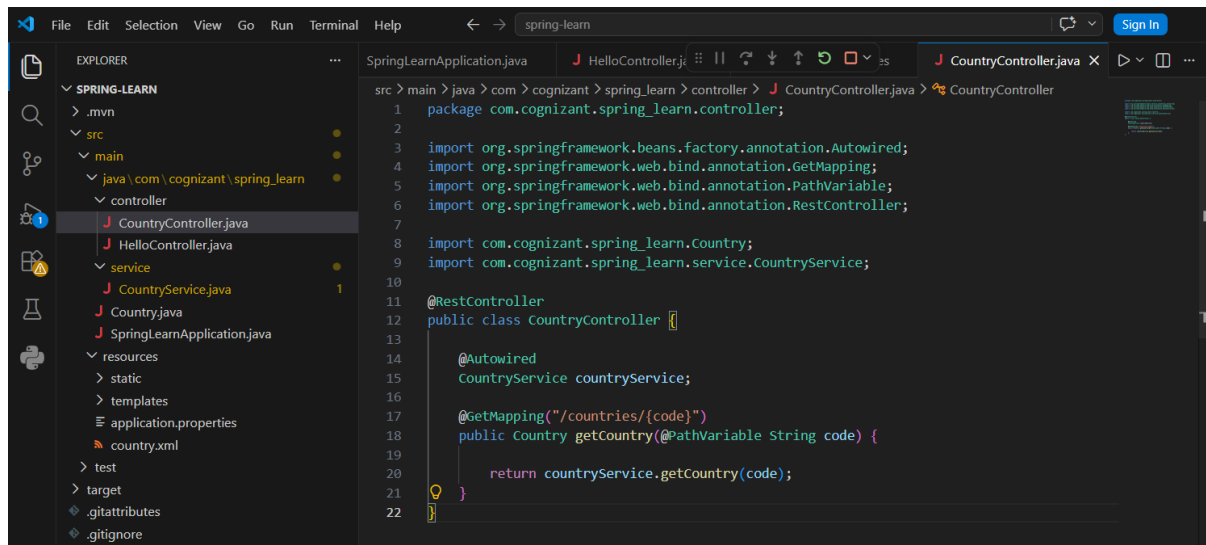
Flows

Humid Now

20:57 04-07-2026

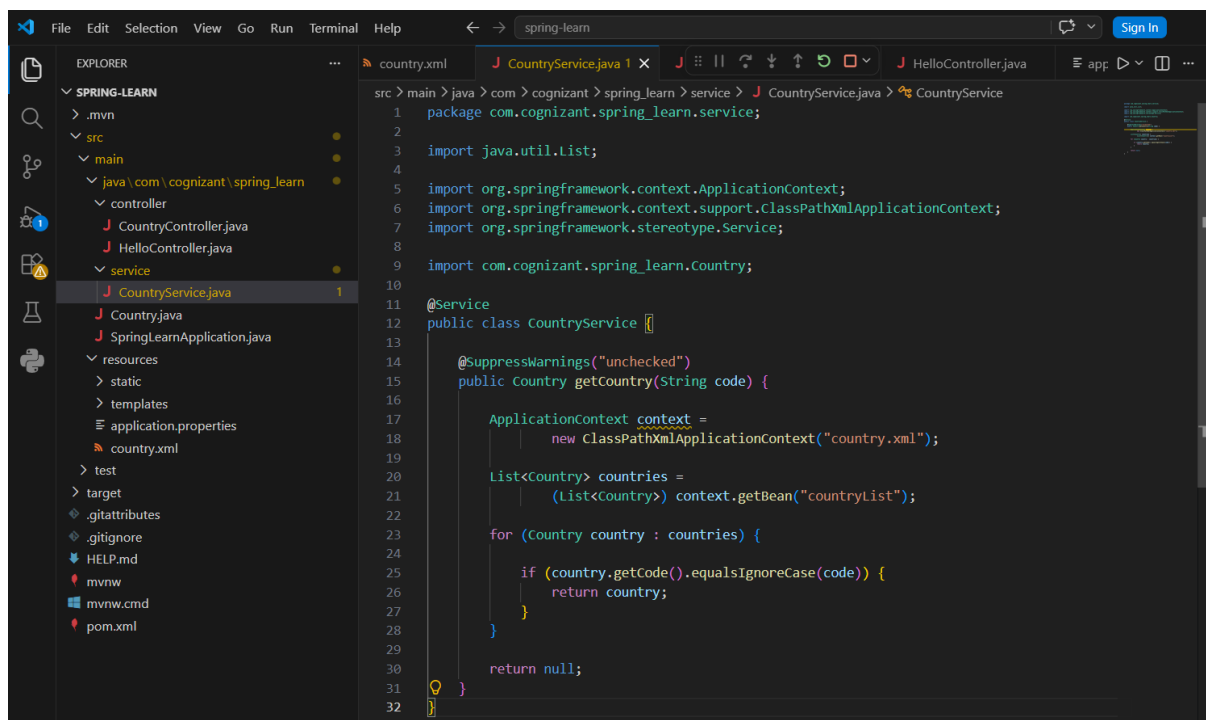
Exercise 4: REST - Get country based on country code

CODE



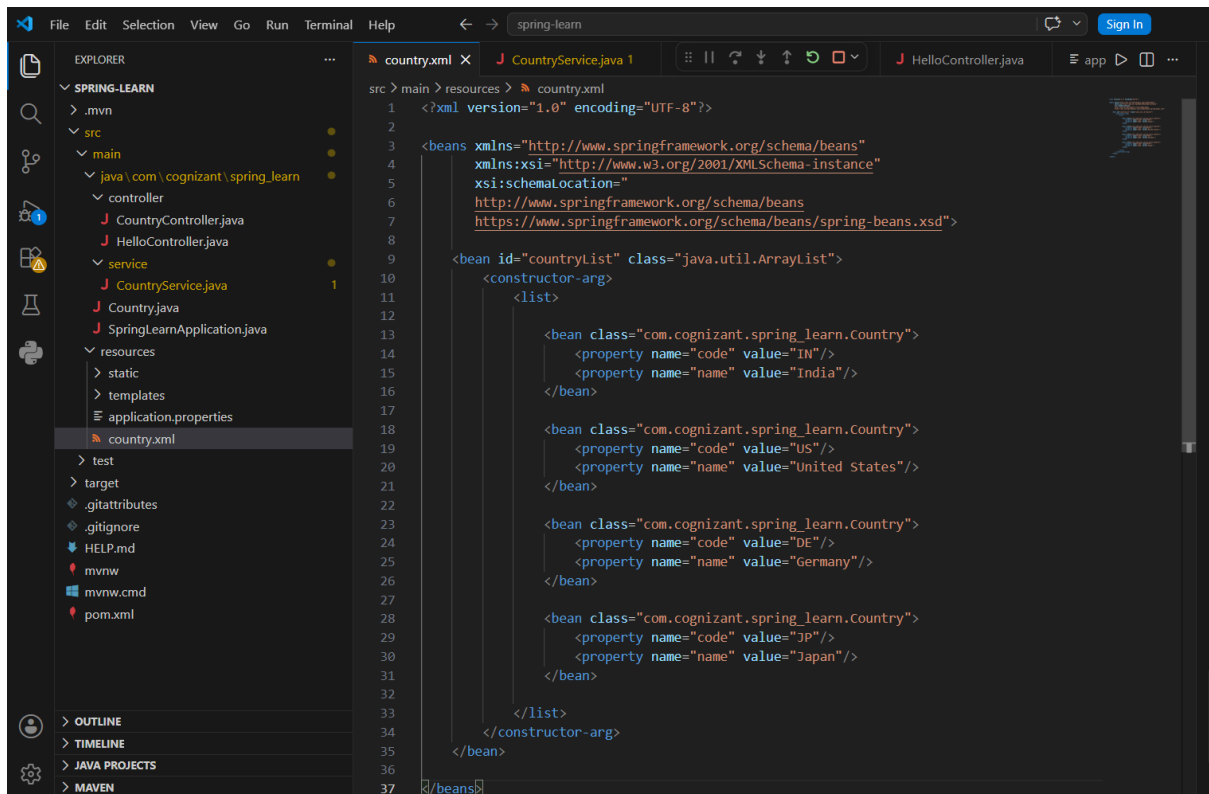
This screenshot shows the VS Code editor with the `CountryController.java` file open. The Explorer on the left shows the project structure with the file selected. The code in the editor is as follows:

```
src > main > java > com > cognizant > spring_learn > controller > CountryController.java > CountryController
1 package com.cognizant.spring_learn.controller;
2
3 import org.springframework.beans.factory.annotation.Autowired;
4 import org.springframework.web.bind.annotation.GetMapping;
5 import org.springframework.web.bind.annotation.PathVariable;
6 import org.springframework.web.bind.annotation.RestController;
7
8 import com.cognizant.spring_learn.Country;
9 import com.cognizant.spring_learn.service.CountryService;
10
11 @RestController
12 public class CountryController {
13
14     @Autowired
15     CountryService countryService;
16
17     @GetMapping("/countries/{code}")
18     public Country getCountry(@PathVariable String code) {
19
20         return countryService.getCountry(code);
21     }
22 }
```



This screenshot shows the VS Code editor with the `CountryService.java` file open. The Explorer on the left shows the project structure with the file selected. The code in the editor is as follows:

```
src > main > java > com > cognizant > spring_learn > service > CountryService.java > CountryService
1 package com.cognizant.spring_learn.service;
2
3 import java.util.List;
4
5 import org.springframework.context.ApplicationContext;
6 import org.springframework.context.support.ClassPathXmlApplicationContext;
7 import org.springframework.stereotype.Service;
8
9 import com.cognizant.spring_learn.Country;
10
11 @Service
12 public class CountryService {
13
14     @SuppressWarnings("unchecked")
15     public Country getCountry(String code) {
16
17         ApplicationContext context =
18             new ClassPathXmlApplicationContext("country.xml");
19
20         List<Country> countries =
21             (List<Country>) context.getBean("countryList");
22
23         for (Country country : countries) {
24
25             if (country.getCode().equalsIgnoreCase(code)) {
26                 return country;
27             }
28         }
29
30         return null;
31     }
32 }
```



OUTPUT

```

Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Khushi Ahi\OneDrive\Desktop\SpringBoot\spring-learn> cmd /C ""C:\Users\Khushi Ahi\.vscode\extensions\redhat.java-1.55.0-win32-x64\jre\21.0.11-win32-x86_64\bin\java.exe" @C:\Users\KHUSHI~1\AppData\Local\Temp\cp_5pojof964mfllwv03wiki2i3f.argfile com.cognizant.spring_learn.SpringLearnApplication ""

  ____ _
 / ___ \ | |
/ /___ \| | |
\___)___|_|_|
=====|_|=====|_|_/___/

:: Spring Boot ::                (v4.1.0)

2026-07-04T21:09:28.075+05:30 INFO 31884 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 21.0.11 with PID 31884 (C:\Users\Khushi Ahi\OneDrive\Desktop\SpringBoot\spring-learn\target\classes started by Khushi Ahi i n C:\Users\Khushi Ahi\OneDrive\Desktop\SpringBoot\spring-learn)
2026-07-04T21:09:28.080+05:30 INFO 31884 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-04T21:09:28.159+05:30 INFO 31884 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-04T21:09:28.159+05:30 INFO 31884 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-04T21:09:29.439+05:30 INFO 31884 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8084 (http)
2026-07-04T21:09:29.468+05:30 INFO 31884 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-04T21:09:29.469+05:30 INFO 31884 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-04T21:09:29.533+05:30 INFO 31884 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1371 ms
2026-07-04T21:09:30.039+05:30 INFO 31884 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8084 (http) with context path '/'
2026-07-04T21:09:30.046+05:30 INFO 31884 --- [spring-learn] [ restartedMain] c.c.spring_learn.

```

```
SpringLearnApplication : Started SpringLearnApplication in 2.683 seconds (process running for 3
.31)
2026-07-04T21:10:05.118+05:30 INFO 31884 --- [spring-learn] [nio-8084-exec-1] o.a.c.c.C.[Tomcat
].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-04T21:10:05.119+05:30 INFO 31884 --- [spring-learn] [nio-8084-exec-1] o.s.web.servlet.D
ispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-04T21:10:05.122+05:30 INFO 31884 --- [spring-learn] [nio-8084-exec-1] o.s.web.servlet.D
ispatcherServlet : Completed initialization in 1 ms
Inside Country Constructor
Inside setCode()
Inside setName()
Inside Country Constructor
Inside setCode()
Inside setName()
Inside Country Constructor
Inside setCode()
Inside setName()
Inside Country Constructor
Inside setCode()
Inside setName()
Inside getCode()
Inside getCode()
Inside getName()
Inside Country Constructor
Inside setCode()
Inside setName()
Inside Country Constructor
Inside setCode()
Inside setName()
```

localhost:8084/countries/de x + Ask Gemini - + x

localhost:8084/countries/de ☆ 📄 School ⋮

Pretty print ☐

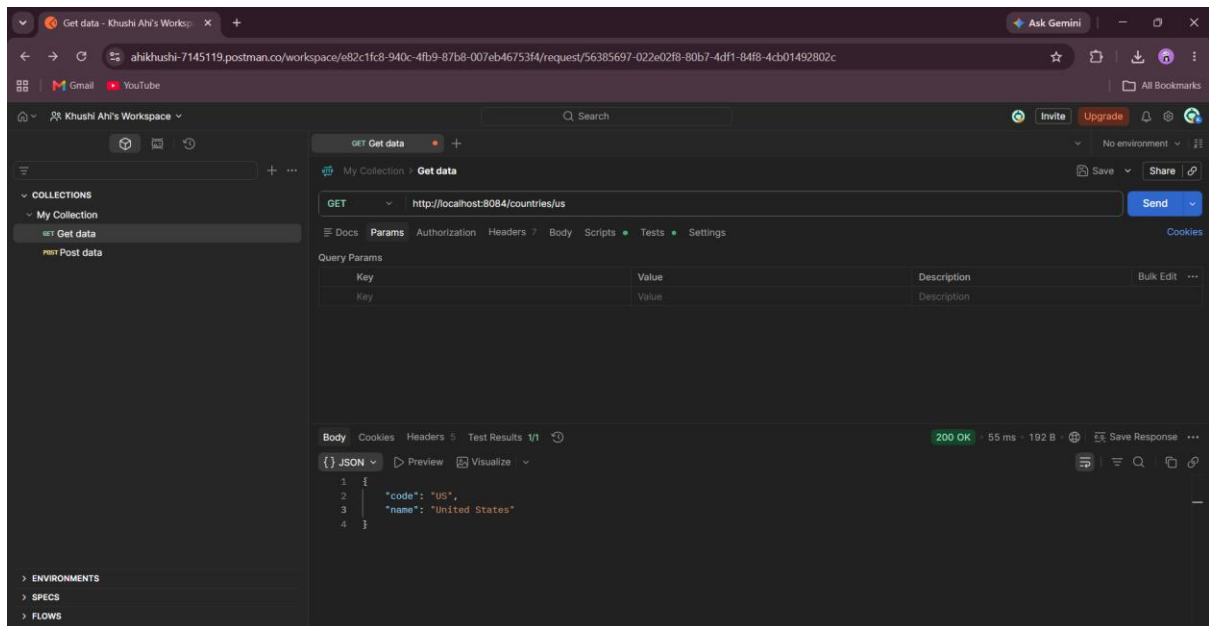
```
{"code": "DE", "name": "Germany"}
```

localhost:8084/countries/in x + Ask Gemini - + x

localhost:8084/countries/in ☆ 📄 School ⋮

Pretty print ☐

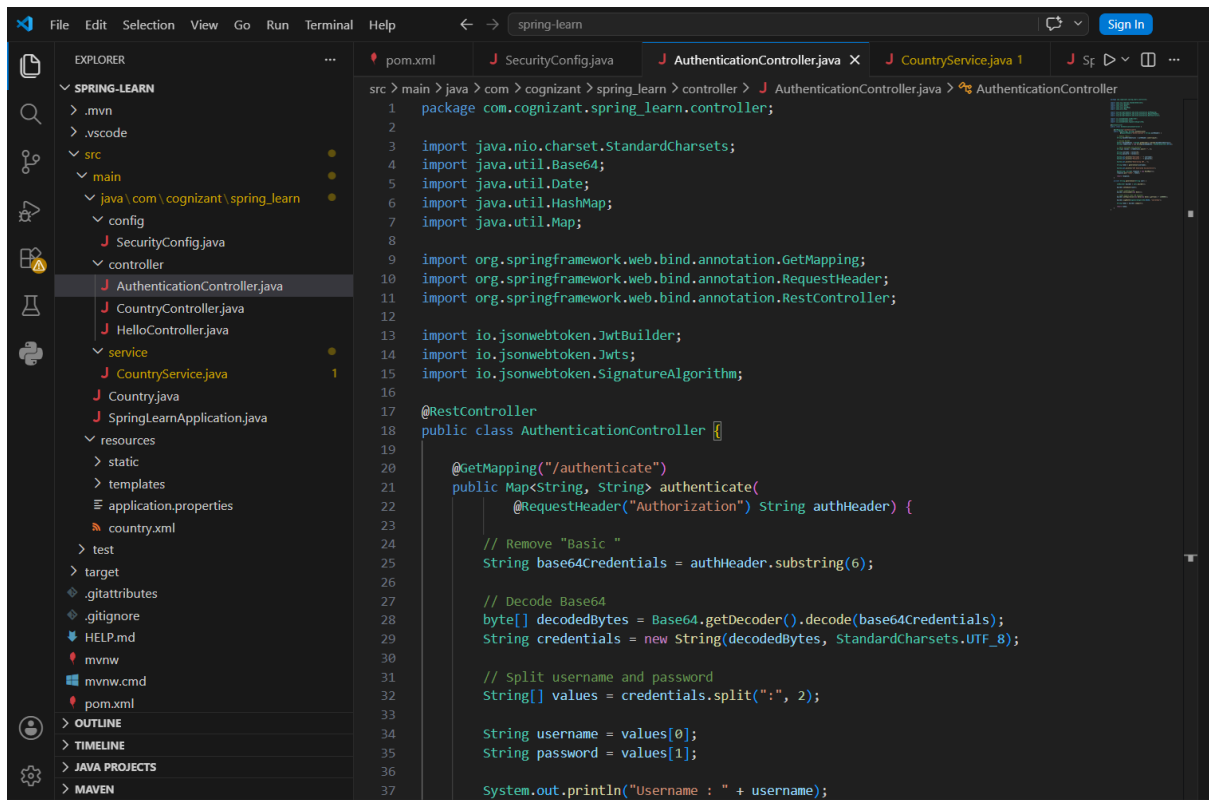
```
{"code": "IN", "name": "India"}
```



File- 5.JWT-handson

Exercise: Create authentication service that returns JWT

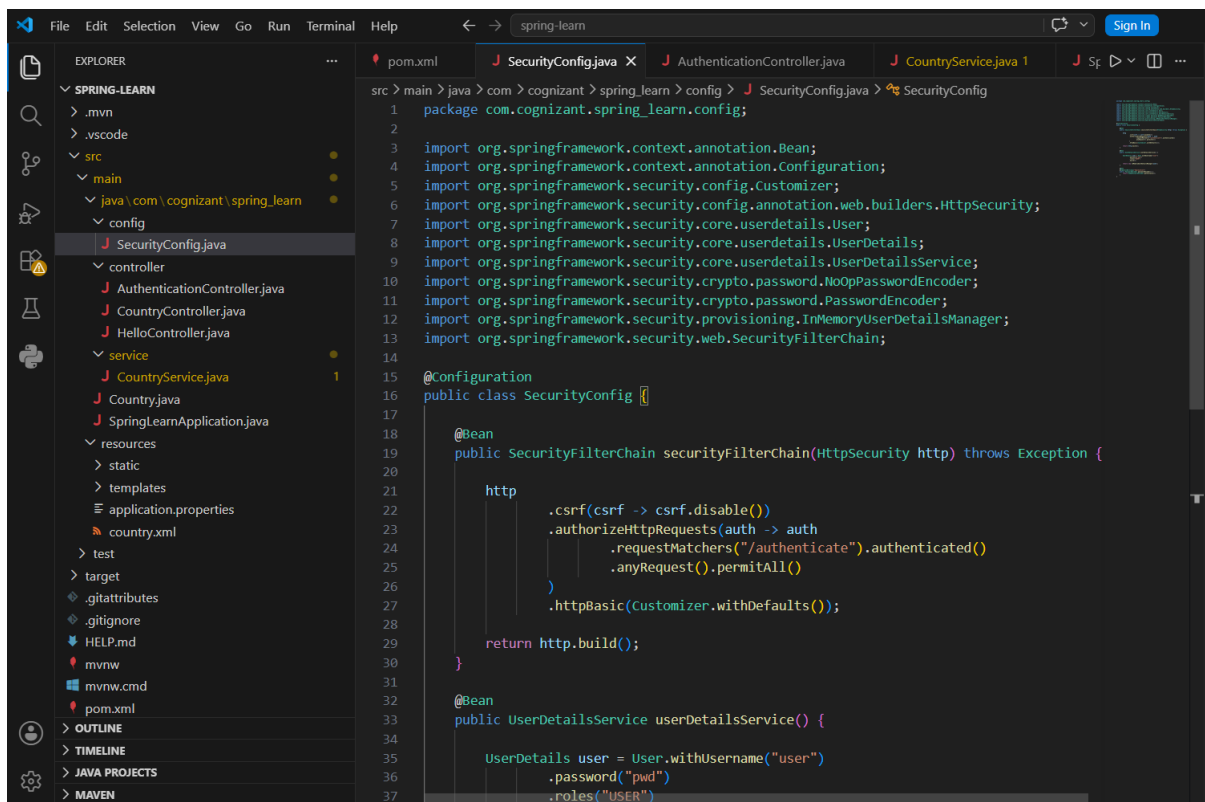
CODE



```

38     System.out.println("Password : " + password);
39
40     System.out.println("Generating JWT...");
41
42     String token = generateJwt(username);
43
44     System.out.println("JWT Generated Successfully");
45
46     Map<String, String> response = new HashMap<>();
47     response.put("token", token);
48
49     return response;
50 }
51
52 private String generateJwt(String user) {
53
54     JwtBuilder builder = Jwts.builder();
55
56     builder.setSubject(user);
57
58     // Token creation time
59     builder.setIssuedAt(new Date());
60
61     // Token expiry time (20 minutes)
62     builder.setExpiration(new Date(new Date().getTime() + 1200000));
63
64     builder.signWith(SignatureAlgorithm.HS256, "secretkey");
65
66     String token = builder.compact();
67
68     return token;
69 }
70 }

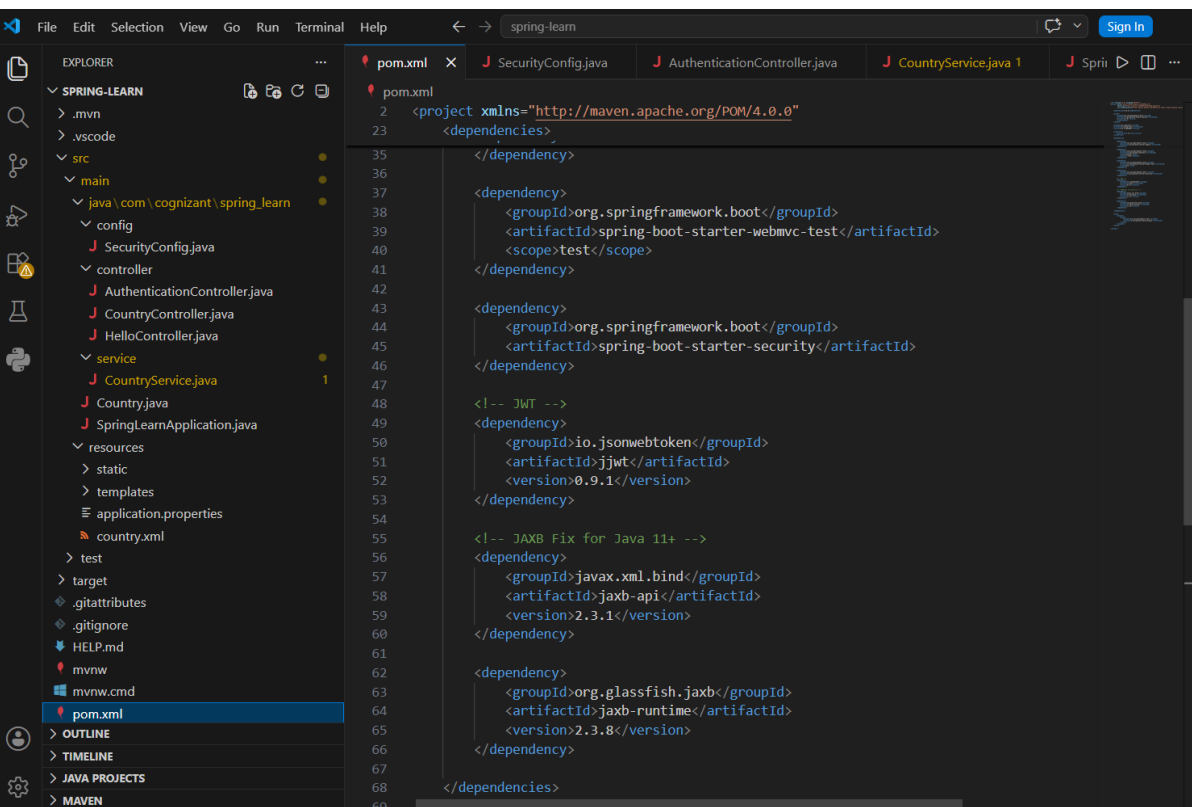
```



```

38         .build();
39
40         return new InMemoryUserDetailsManager(user);
41     }
42
43     @Bean
44     @SuppressWarnings("deprecation")
45     public PasswordEncoder passwordEncoder() {
46         return NoOpPasswordEncoder.getInstance();
47     }
48 }

```



OUTPUT

localhost:8084/authenticate

Pretty print ☐

```
{"token":""}
```

